

Calculus and Linear Algebra

# Sets, Real Numbers and Mathematical Language

Lecture 1

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16 September 2026

# Before we start: what is this course really about?

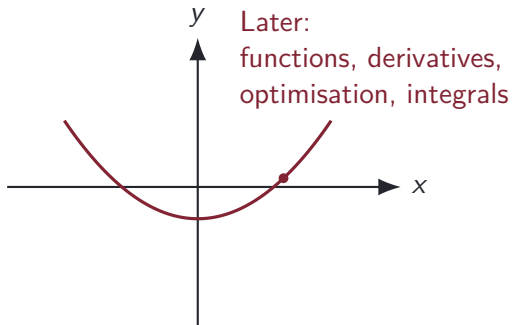
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## Learning to read mathematical objects

A formula is not a decoration. It tells us:

- ▶ what objects we are working with;
- ▶ which operations are allowed;
- ▶ what must be true before we calculate;
- ▶ what a result means geometrically.

Today: build the language we will use for the whole course.



# Today's route

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## The point

By the end of the lesson you should be able to translate quickly between **words**, **symbols**, and a **picture on the number line**.

Warm-up: read the symbols before calculating

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What do these statements say in ordinary English?

1.  $3 \in \mathbb{N}$

2.  $-2 \notin \mathbb{N}$

3.  $\{1, 2\} \subset \{1, 2, 3\}$

4.  $x \in [-1, 4)$

5.  $A \cap B = \emptyset$

6.  $|x - 2| < 1$

**Your turn** Do not solve anything yet. Just say what each line means.

# A set is a collection of objects

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A **set** is a collection whose elements are clearly determined.

$$A = \{1, 3, 5, 7\}$$

We write

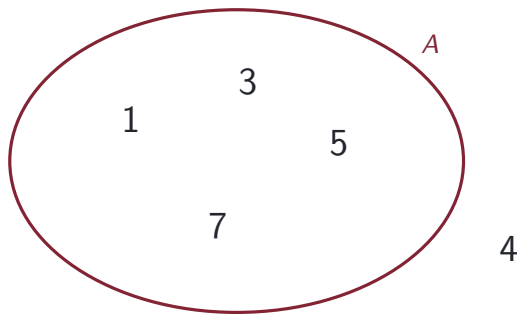
$$3 \in A, \quad 4 \notin A.$$

Order does not matter:

$$\{1, 3, 5\} = \{5, 1, 3\}.$$

Repeated elements do not matter:

$$\{1, 1, 3\} = \{1, 3\}.$$



## Two common ways to describe a set

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### Roster form

List the elements:

$$A = \{2, 4, 6, 8\}.$$

### Set-builder form

Describe the property:

$$A = \{x \in \mathbb{N} : 2 \leq x \leq 8, x \text{ even}\}.$$

Good when the set is small and explicit.

Good when the rule matters more than the list.

**Your turn** Write  $\{-3, -2, -1, 0, 1, 2, 3\}$  in set-builder notation.

## Exercise: translate into symbols

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Write each sentence in mathematical notation.

1.  $x$  belongs to the set of real numbers.
2. 5 is not an element of the set  $A$ .
3.  $B$  contains the elements  $-1, 0, 1$ .
4.  $A$  contains only positive integers smaller than 6.

Notation is a language. Fluency comes from translating both ways.

## Subsets: a set inside another set

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If every element of  $A$  also belongs to  $B$ ,  
then

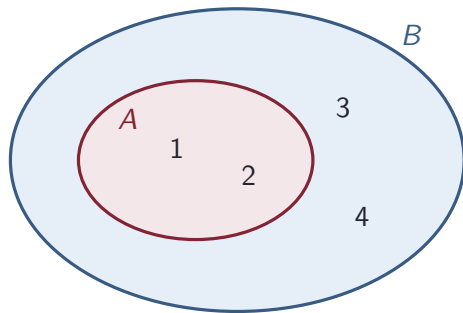
$$A \subseteq B.$$

Example:

$$A = \{1, 2\}, \quad B = \{1, 2, 3, 4\}.$$

Hence  $A \subseteq B$ . **Important:** an element  
and a subset are different objects.

$$1 \in B, \quad \{1\} \subseteq B.$$



## A common trap

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Let  $A = \{1, 2, 3\}$ . Which statements are true?

$$1 \in A$$

$$\{1\} \in A$$

$$\{1\} \subseteq A$$

$$A \subseteq A$$

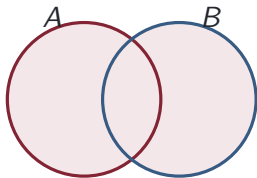
**Your turn** Vote first. Then justify your answer with words, not only symbols.

# Union and intersection

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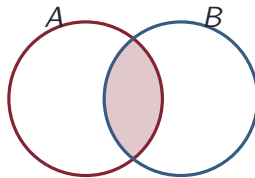
**Union:** everything in  $A$  or  $B$  (or both)

$$A \cup B.$$



**Intersection:** everything in both  $A$  and  $B$

$$A \cap B.$$



# Work from a concrete example

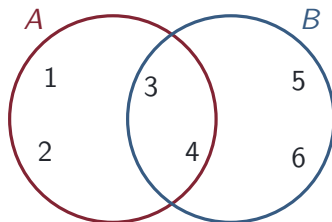
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Let

$$A = \{1, 2, 3, 4\}, \quad B = \{3, 4, 5, 6\}.$$

Find:

$$A \cup B, \quad A \cap B, \quad A \setminus B, \quad B \setminus A.$$



# Empty set and disjoint sets

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The **empty set** has no elements:

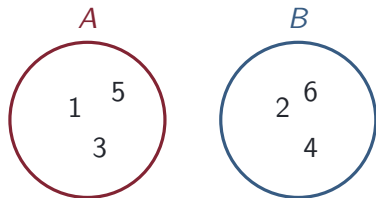
$$\emptyset.$$

Two sets are **disjoint** when they have no common element:

$$A \cap B = \emptyset.$$

Example:

$$\{1, 3, 5\} \cap \{2, 4, 6\} = \emptyset.$$



# Checkpoint 1

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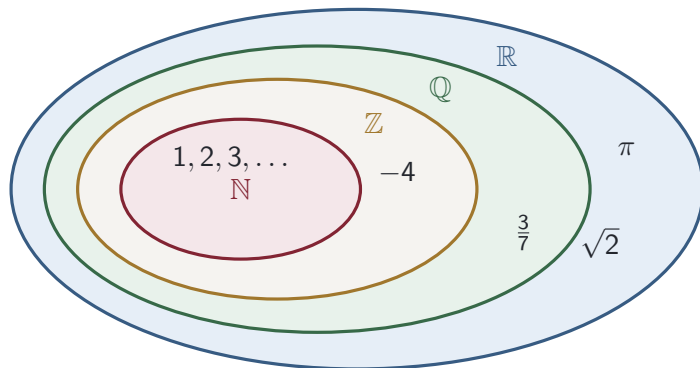
Let  $A = \{x \in \mathbb{Z} : -2 \leq x \leq 3\}$  and  $B = \{-4, -2, 0, 2, 4\}$ .

1. Write  $A$  in roster form.
2. Find  $A \cap B$ .
3. Find  $A \cup B$ .
4. Is  $\{-2, 0, 2\} \subseteq A$ ?
5. Is  $4 \in A \cup B$ ?

3 minutes – compare answers with the person next to you.

# The number system: nested sets

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$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}.$$

# Natural numbers and integers

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## Natural numbers

$$\mathbb{N} = \{1, 2, 3, \dots\}$$

Counting numbers.

Some books include 0 in  $\mathbb{N}$ . Always check the convention.

## Integers

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$$

Positive and negative whole numbers, including zero.

**Your turn** Which set is the smallest one containing each number?

7,      -5,      0.

# Rational numbers

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A number is **rational** if it can be written as a ratio of integers:

$$\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0 \right\}.$$

Examples:

$$\frac{2}{3}, \quad -\frac{11}{4}, \quad 5 = \frac{5}{1}, \quad 0.125 = \frac{1}{8}.$$

A decimal expansion of a rational number either **terminates** or **eventually repeats**.

$$\frac{1}{3} = 0.333\dots, \quad \frac{7}{12} = 0.58333\dots$$

# Irrational numbers: real, but not rational

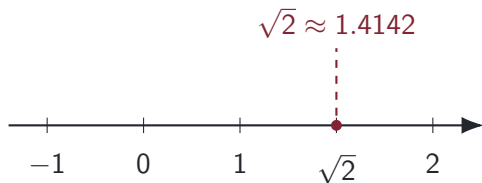
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Numbers such as

$$\sqrt{2}, \quad \pi, \quad e$$

cannot be written as  $p/q$  with integers  $p, q$ .  
They are **irrational**. Together,

$$\text{rationals} + \text{irrationals} = \mathbb{R}.$$



## Your turn: classify the numbers

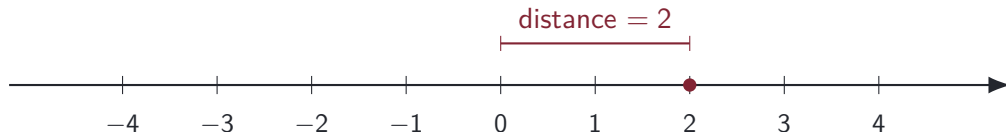
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For each number, choose the **smallest** set among  $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$  that contains it.

| Number        | $\mathbb{N}$ | $\mathbb{Z}$ | $\mathbb{Q}$ | $\mathbb{R}$ |
|---------------|--------------|--------------|--------------|--------------|
| 12            |              |              |              |              |
| -3            |              |              |              |              |
| $\frac{7}{5}$ |              |              |              |              |
| 0.4           |              |              |              |              |
| $\sqrt{9}$    |              |              |              |              |
| $\sqrt{5}$    |              |              |              |              |

# The real line: numbers become geometry

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## Key idea

The order relation  $<$  gives the line a direction. Distance turns algebra into geometry.

**Your turn** Where would you place  $-2.5$ ,  $1/3$ , and  $\sqrt{2}$ ?

# Intervals: a compact way to describe many numbers

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## Closed interval

$$[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$$



## Open interval

$$(a, b) = \{x \in \mathbb{R} : a < x < b\}$$



Square bracket = endpoint included. Round bracket = endpoint excluded.

# Half-open intervals

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$$[a, b) = \{x : a \leq x < b\}$$



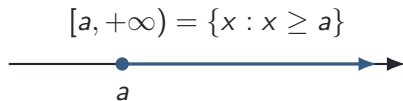
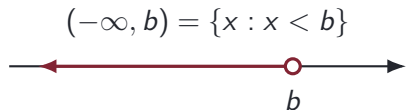
$$(a, b] = \{x : a < x \leq b\}$$



**Your turn** Draw  $[-2, 3)$  on a number line.

# Unbounded intervals

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$\pm\infty$  are not real numbers, so we never use square brackets with them.

# Fast translation drill

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Translate each interval into an inequality, then sketch it.

1.  $[-3, 2]$

2.  $(1, 5)$

3.  $(-\infty, 4]$

4.  $[-1, +\infty)$

5.  $(-2, 0]$

6.  $\mathbb{R}$

Goal: interval notation should become automatic.

## And now the reverse translation

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Write each inequality using interval notation.

1.  $-2 \leq x < 4$

2.  $x > 3$

3.  $x \leq -1$

4.  $0 < x \leq 7$

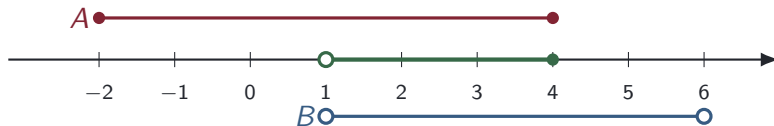
**Your turn** For each one, mark endpoints with open/closed circles before writing the interval.

## Intervals are sets: we can intersect them

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Find

$$[-2, 4] \cap (1, 6).$$



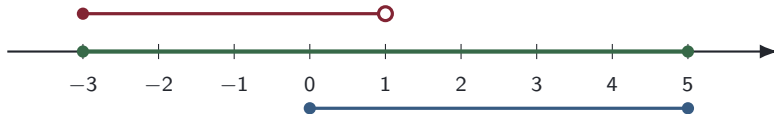
$$[-2, 4] \cap (1, 6) = \boxed{(1, 4]}.$$

# Intervals are sets: and we can take unions

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Find

$$[-3, 1) \cup [0, 5].$$



$$[-3, 1) \cup [0, 5] = \boxed{[-3, 5]}.$$

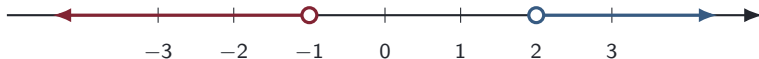
The two intervals overlap, so the union is one continuous interval.

What if there is a gap?

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Find

$$(-\infty, -1) \cup (2, +\infty).$$



Can this union be written as a single interval? Why not?

## Inequalities: think on the line

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Consider

$$2x - 3 < 5.$$

Solving is not just algebra; it identifies a set of real numbers:

$$2x < 8 \implies x < 4.$$



$$\boxed{(-\infty, 4)}.$$

## One rule you must not forget

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When you multiply or divide an inequality by a **negative number**, the inequality sign reverses.

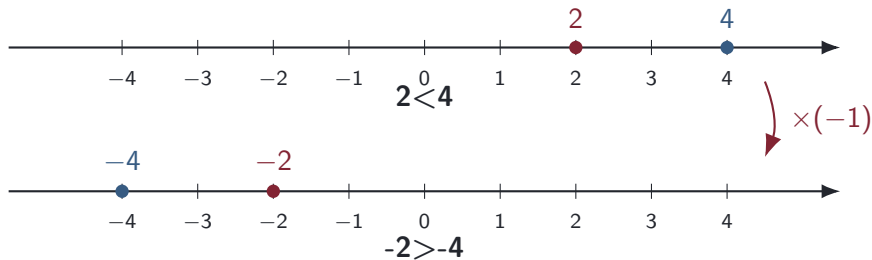
$$-2x > 6$$

↓ divide by  $-2$

$$\boxed{x < -3}.$$

Why? Because multiplication by  $-1$  reflects the number line around zero.

## See the sign reversal geometrically



## Exercise set: linear inequalities

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Solve and write the solution in interval notation.

1.  $3x + 1 \leq 10$

2.  $5 - 2x > 1$

3.  $4x - 7 \geq 2x + 3$

4.  $2(1 - x) < 3x + 7$

5.  $-3 \leq 2x + 1 < 7$

For no. 5, treat the double inequality as two conditions at once.

# Double inequalities

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Solve

$$-3 \leq 2x + 1 < 7.$$

We perform the same operation on all three parts:

$$-4 \leq 2x < 6$$

$$-2 \leq x < 3.$$



$$\boxed{[-2, 3)}.$$

# Absolute value is distance from zero

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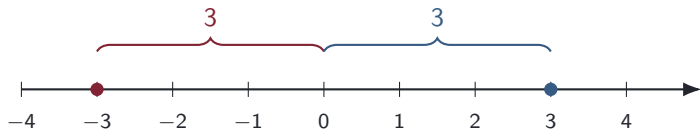
$|x|$  = distance of  $x$  from 0.

Therefore

$$|3| = 3, \quad |-3| = 3.$$

Distance is never negative:

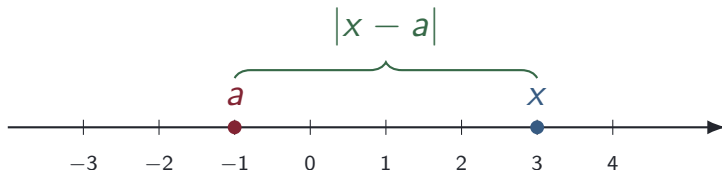
$$|x| \geq 0.$$



# Absolute value between two points

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$|x - a|$  = distance between  $x$  and  $a$ .



This interpretation will reappear in limits, continuity, approximation and optimisation.

## Absolute-value inequalities: start from the picture

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What does

$$|x| < 2$$

mean? It says: *the distance of  $x$  from zero is less than 2.*



$$-2 < x < 2$$

$\iff$

$$x \in (-2, 2).$$

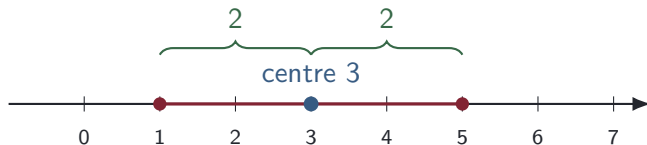
## Shift the centre

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Solve

$$|x - 3| \leq 2.$$

Interpretation:  $x$  must be at distance at most 2 from the centre 3.



$$\boxed{1 \leq x \leq 5}.$$

## General rule: inside a radius

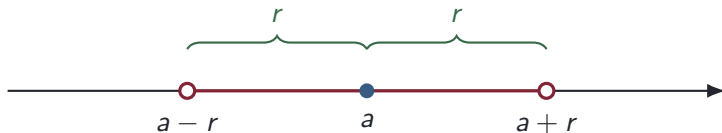
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For  $r > 0$ ,

$$|x - a| < r$$

means that  $x$  lies **inside** the interval of radius  $r$  centred at  $a$ :

$$a - r < x < a + r.$$



## General rule: outside a radius

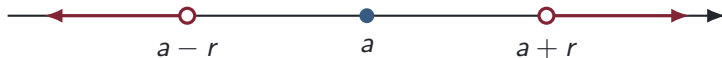
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For  $r > 0$ ,

$$|x - a| > r$$

means that  $x$  lies **outside** the interval of radius  $r$  centred at  $a$ :

$$x < a - r \quad \text{or} \quad x > a + r.$$



## Exercise set: absolute value

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Solve and represent the solution on the real line.

1.  $|x| < 4$

2.  $|x| \geq 3$

3.  $|x - 2| < 1$

4.  $|x + 1| \leq 2$

5.  $|2x - 1| < 5$

Always ask first: what is the centre? what is the radius?

# From algebra to geometry: $|2x - 1| < 5$

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$$|2x - 1| < 5$$

$$-5 < 2x - 1 < 5$$

$$-4 < 2x < 6$$

$$\boxed{-2 < x < 3}.$$



## A useful application: tolerance

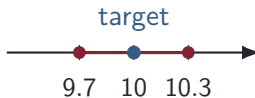
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Suppose a measurement is intended to be 10, with tolerance 0.3. “Acceptable” means

$$|x - 10| \leq 0.3.$$

Therefore

$$9.7 \leq x \leq 10.3.$$



# A preview of functions: domain restrictions come from sets

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Soon we will study functions such as

$$f(x) = \frac{1}{x-2}, \quad g(x) = \sqrt{x+1}, \quad h(x) = \log x.$$

Before doing anything else, we must ask:

**For which real numbers is the formula meaningful?**

$$x \neq 2$$

$$x \geq -1$$

$$x > 0$$

Today's set and interval notation will become the language of **domains**.

## Challenge: describe the domain as a set

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Write the natural domain of each expression in interval notation.

1.  $f(x) = \frac{1}{x+4}$

2.  $g(x) = \sqrt{5-x}$

3.  $h(x) = \frac{1}{\sqrt{x-1}}$

This is only a preview. The point is to use intervals correctly.

# Exercises

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1. Write  $A = \{x \in \mathbb{Z} : -1 < x \leq 4\}$  in roster form.
2. Compute  $[-2, 3) \cap [1, 5]$ .
3. Compute  $(-\infty, 0] \cup (-1, 2)$ .
4. Solve  $3 - 2x \leq 7$ .
5. Solve  $-1 < 2x + 3 \leq 7$ .
6. Solve  $|x - 4| < 2$ .
7. Solve  $|x + 2| \geq 3$ .
8. Classify  $-7/2$ ,  $\sqrt{16}$  and  $\pi$ .

8 minutes – write full notation, not only final numbers.

## Quick solutions

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1.  $A = \{0, 1, 2, 3, 4\}$

2.  $[1, 3)$

3.  $(-\infty, 2)$

4.  $x \geq -2 \Rightarrow [-2, +\infty)$

5.  $-2 < x \leq 2 \Rightarrow (-2, 2]$

6.  $2 < x < 6 \Rightarrow (2, 6)$

7.  $x \leq -5$  or  $x \geq 1$

8.  $-7/2 \in \mathbb{Q}$ ,  $\sqrt{16} = 4 \in \mathbb{N}$ ,  $\pi \in \mathbb{R} \setminus \mathbb{Q}$

**Check** Compare your reasoning, not only the answers.

# Five ideas to leave with today

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1. A set is a collection of well-defined objects.
2.  $\in$  relates an **element** to a set;  $\subseteq$  relates one **set** to another.
3.  $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$ .
4. Intervals are sets of real numbers described by inequalities.
5. Absolute value is distance:  $|x - a|$  measures how far  $x$  is from  $a$ .

The real line should now be your default mental picture.

## Exit ticket

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Before you leave, answer these three questions without notes.

1. What is the difference between  $2 \in A$  and  $\{2\} \subseteq A$ ?
2. Write  $-1 \leq x < 3$  as an interval.
3. Solve  $|x - 5| < 2$  without expanding anything first.

Next lecture: **Vectors – geometry, coordinates and operations.**

**Mathematics becomes easier  
when the picture and the  
symbols  
say the same thing.**

See you on Friday.

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